

Overview

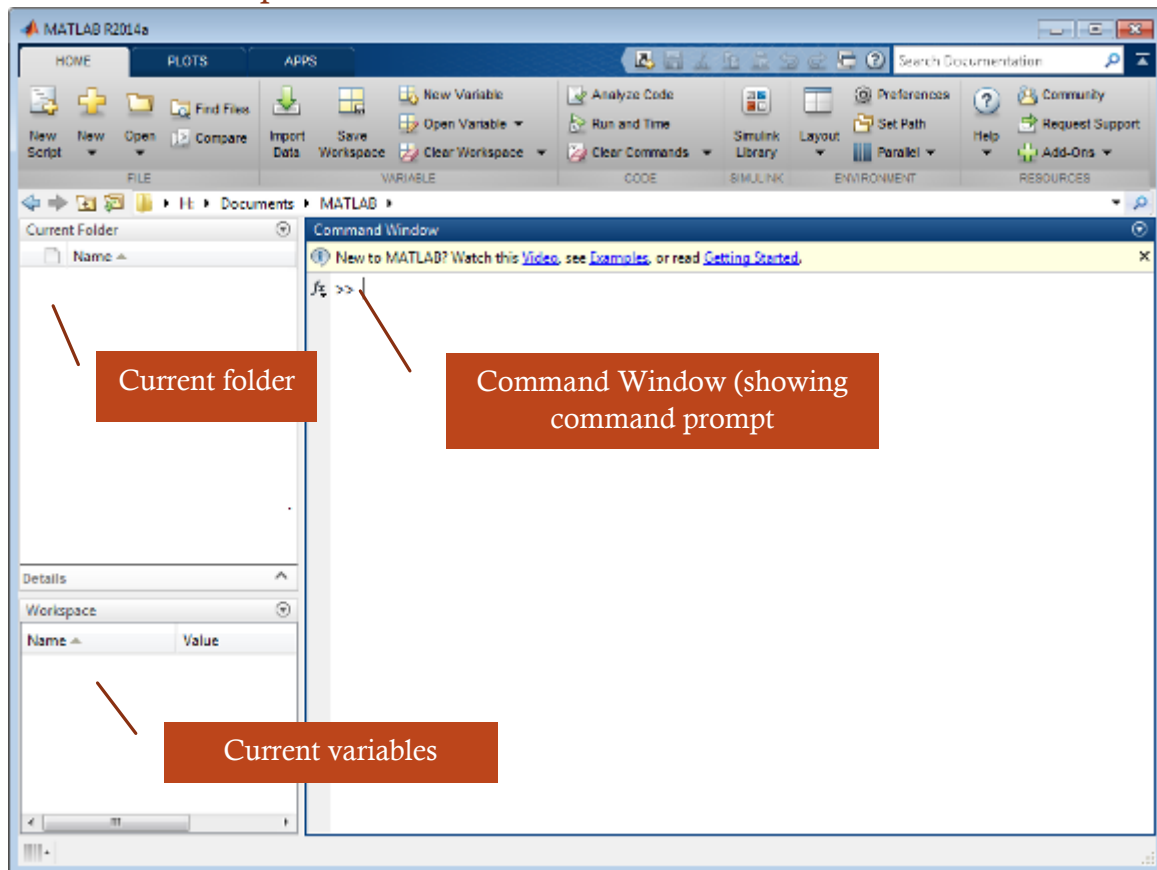
Although this course has a general name of computer-aided design (CAD) it is actually a MATLAB course. You will learn how to use MATLAB as an integrated solution environment and as a programming language. During the first semester, we will focus mainly on how to use MATLAB in general. By the end of it, you will be able to write m-file scripts to automate tasks, load, analyse and save data efficiently and fluently. Whereas in the second semester, you will apply the knowledge you gained previously to various interesting problems in Control system engineering, digital signal, and image, processing and communications systems.

Why use MATLAB

Matlab is a software development environment. It has thousands of ready-to-use tools and algorithms in areas like control, physics, DSP, image processing, communication and many other engineering and scientific disciplines. Its desktop environment lets you work interactively with data with both ease and competence. For me, Matlab removes the need to think about how to program a solution and replace it with how to find a solution to my engineering problem. For example, rather than thinking about how to program subroutine to invert a matrix to use it in a state-space represented control system, I can simply use Matlab powerful commands to calculate the results with a single command. I can even represent my system and find its response with no more than a few lines of code.

Nevertheless, Matlab has also some disadvantages. For starters, it uses large amount of memory which makes it hard to use on a low-end budget computer. You cannot use Matlab for real-time systems because it works under the supervisory of Windows, which throttles the CPU and time schedule in a way that it would not guarantee response within some designated time window. Finally, it is an interpreted language, which often leads to slower execution time than compiled languages such as C. On the other hand, interpreted languages allows one to quickly debug the code line by line very easily.

The Matlab desktop¹



The desktop includes these panels:

- **Current Folder** — Access your files.
- **Command Window** — Enter commands at the command line, indicated by the prompt (`>>`).
- **Workspace** — Explore data that you create or import from files.

As you work in MATLAB, you issue commands that create variables and call functions. For example, create a variable named `a` by typing this statement at the command line:

```
a = 1
```

MATLAB adds variable `a` to the workspace and displays the result in the Command Window.

```
a =
```

```
1
```

¹ This section is taken from the Matlab help file available on https://www.mathworks.com/help/matlab/learn_matlab/desktop.html (accessed 18.10.2014)

Create a few more variables.

```
b = 2
```

```
b =
```

```
2
```

```
c = a + b
```

```
c =
```

```
3
```

```
d = cos(a)
```

```
d =
```

```
0.5403
```

When you do not specify an output variable, MATLAB uses the variable `ans`, short for *answer*, to store the results of your calculation.

```
sin(a)
```

```
ans =
```

```
0.8415
```

If you end a statement with a semicolon, MATLAB performs the computation, but suppresses the display of output in the Command Window.

```
e = a*b;
```

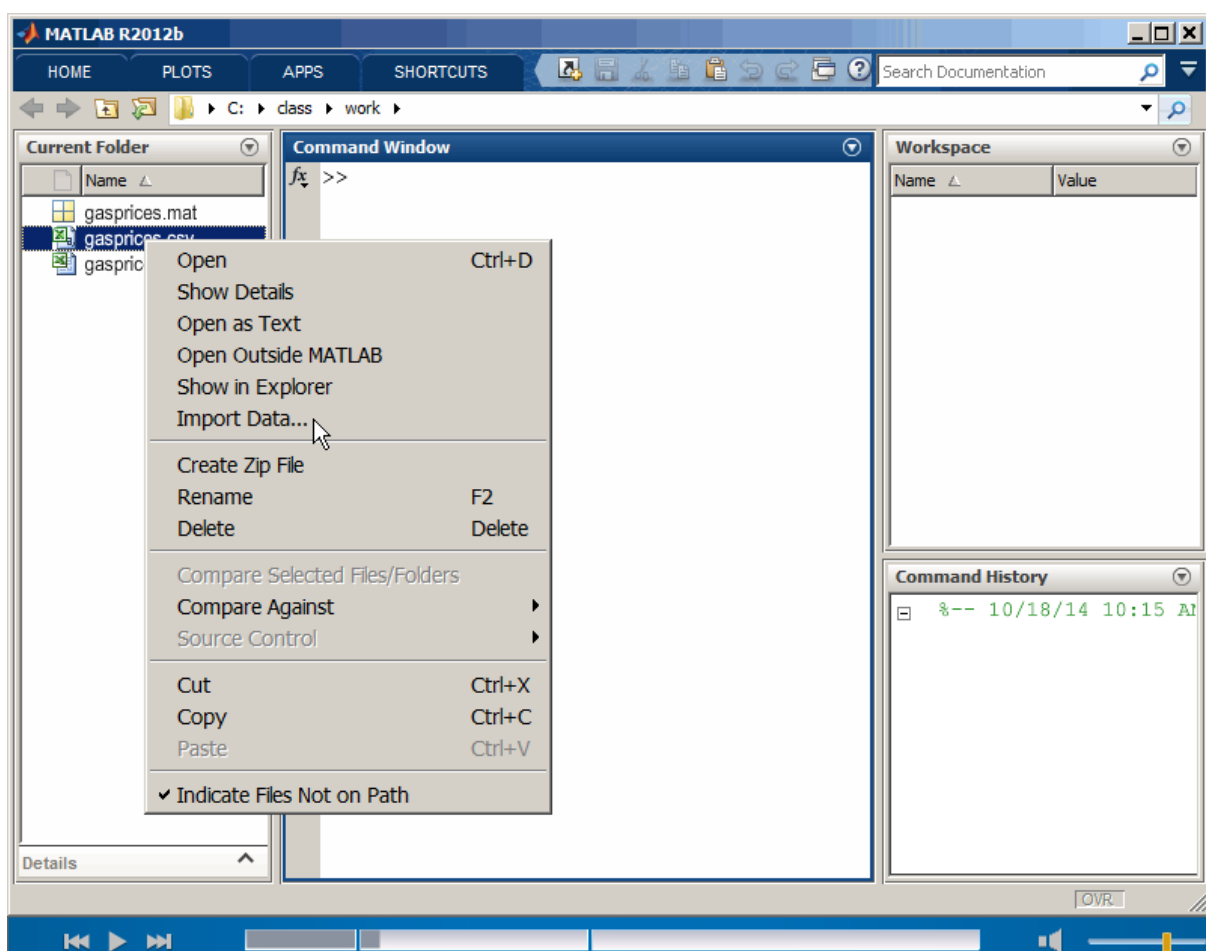
You can recall previous commands by pressing the up- and down-arrow keys, $\uparrow$ and $\downarrow$. Press the arrow keys either at an empty command line or after you type the first few characters of a command. For example, to recall the command `b = 2`, type `b`, and then press the up-arrow key.

Importing data into Matlab

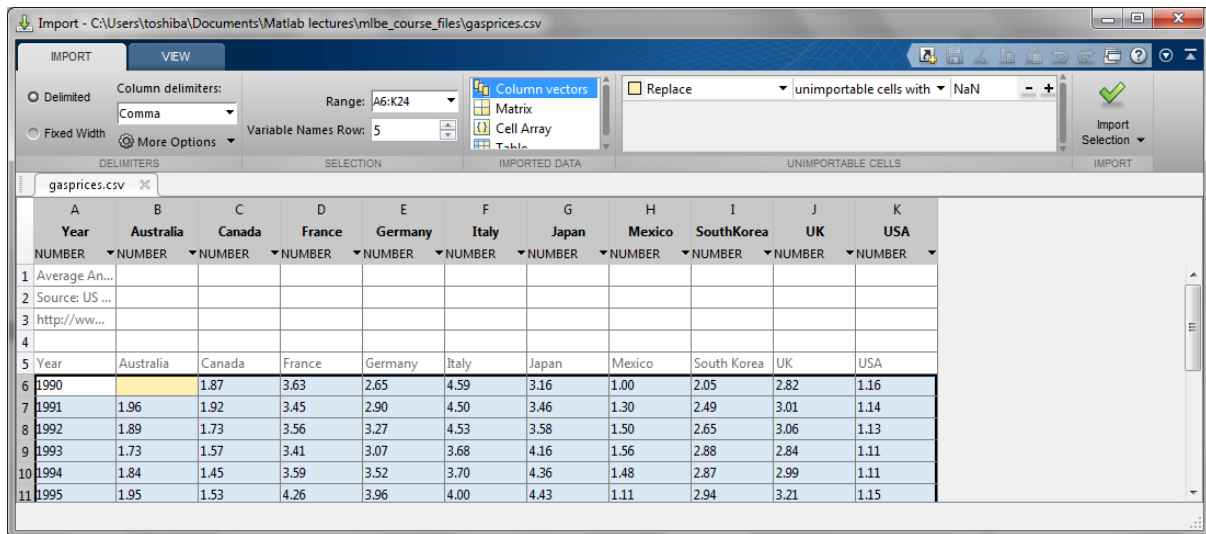
In this lecture, you will learn how to use Matlab GUI to import, analyse and save data. The dataset we will be using is a comma separated values matrix that represents the gas prices of several countries from 1990 to 2008. You should find the file in your course files folder under the name: *gasprices.csv*.

Start by creating an arbitrary folder and coping the file *gasprices.csv* to it. Now switch to Matlab and use its browsing bar to navigate to the folder you just created. Remember that you should always navigate to where your code/data is for it to become the active directory that Matlab can run command on and execute scripts from.

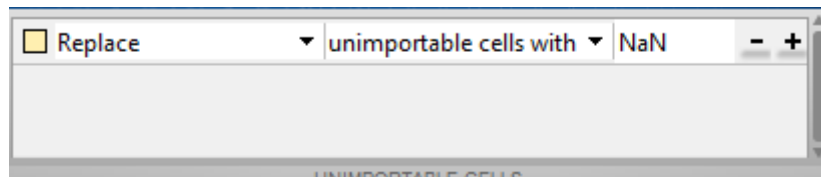
Now, in the current folder pane, right click *gasprices.csv* and choose *open as text*. This will open Matlab internal text viewer so you can take a quick look at the internal organization of data in a file. This step is very important because many real life data set are not as clean and/or complete as one hopes! In such cases, you need to think ahead about how to fill in the gaps and remove clutter and any other unusable parts.



Right click the file again and choose import data. See the import data window



This window shows you how Matlab have interpreted the contents of your file. It automatically detected that the data delimiter is a comma, which is what we want. In addition, the data is considered column based, which Matlab calls Column vector. This means Matlab will generates a vector that holds the years, and 10 other vectors that contains the gas prices of each country. Note that the 1990 gas price of Australia is missing. Matlab offers to replace the empty value with a *NaN*, which stands for *Not a Number*.

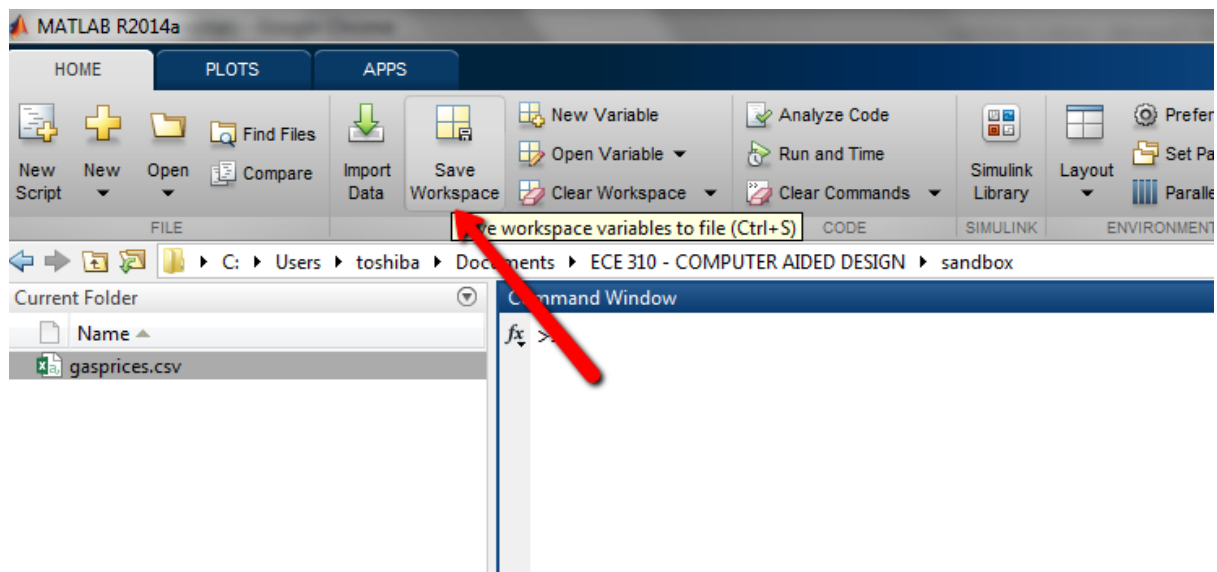


Replace NaN with zero to tell Matlab to use 0 instead. Then click on import selection. Close the import window and return to Matlab desktop. See the imported vectors in the workspace pane.

Workspace		
Name	Value	M
Australia	19x1 double	Na
Canada	19x1 double	1.3
France	19x1 double	3.4
Germany	19x1 double	2.6
Italy	19x1 double	3.5
Japan	19x1 double	2.8
Mexico	19x1 double	1
prices	19x11 double	Na
SouthKorea	19x1 double	2.0
UK	19x1 double	2.8
USA	19x1 double	1.0
Year	19x1 double	19

Saving and loading data

To save your current variables, click save workspace in the home tab of the Matlab desktop window.



by default, the variables will be saved to a file called *matlab.mat*. The mat files is the format that Matlab uses to store data in binary form. Alternatively, you can type:

```
save
```

to save the current workspace to *matlab.mat*. You can also specify the file name by typing it after the save command and even specify which variables you want saved:

```
save prices.mat
```

```
save prices.mat UK USA Year
```

to clear the current workspace of variables, click on clear workspace or type

```
clear
```

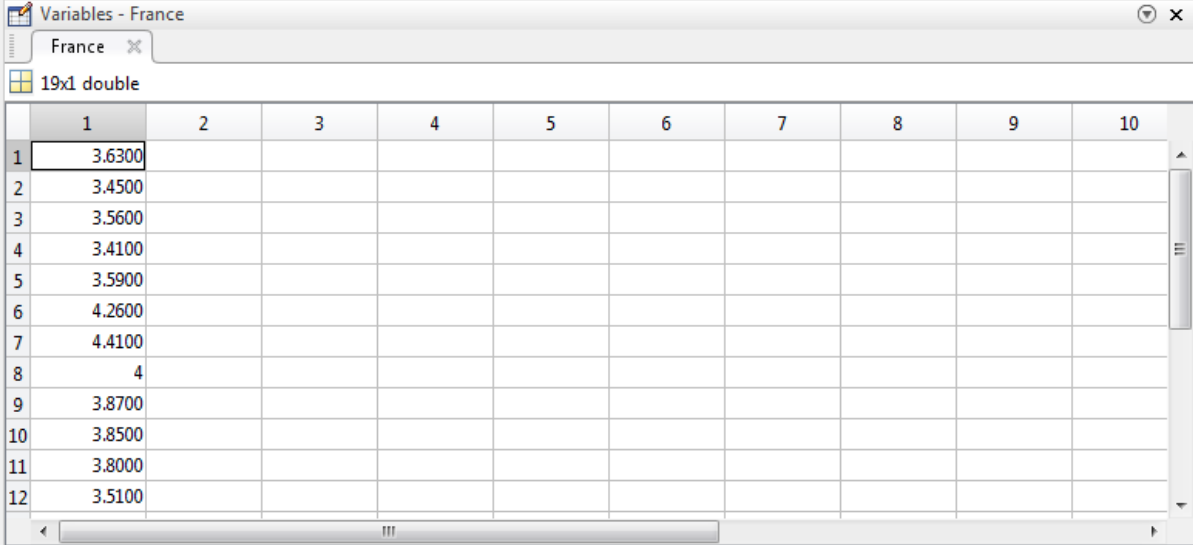
if you want to remove a specific variable, just type its name after the clear command, e.g:

```
clear UK
```

To load data stored in Matlab *mat* format, just double click the file in the current folder pane, or type load followed by the file name

```
load matlab.mat
```

to view the contents of a variable, double click the variable in the workspace pane.

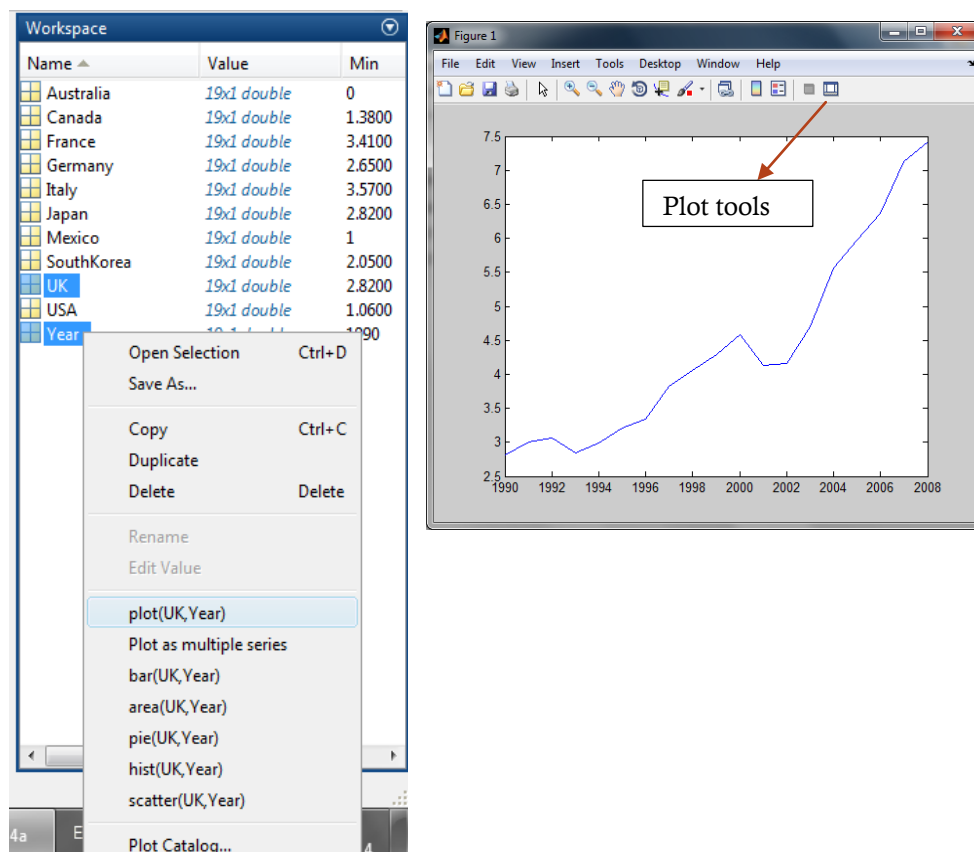


	1	2	3	4	5	6	7	8	9	10
1	3.6300									
2	3.4500									
3	3.5600									
4	3.4100									
5	3.5900									
6	4.2600									
7	4.4100									
8	4									
9	3.8700									
10	3.8500									
11	3.8000									
12	3.5100									

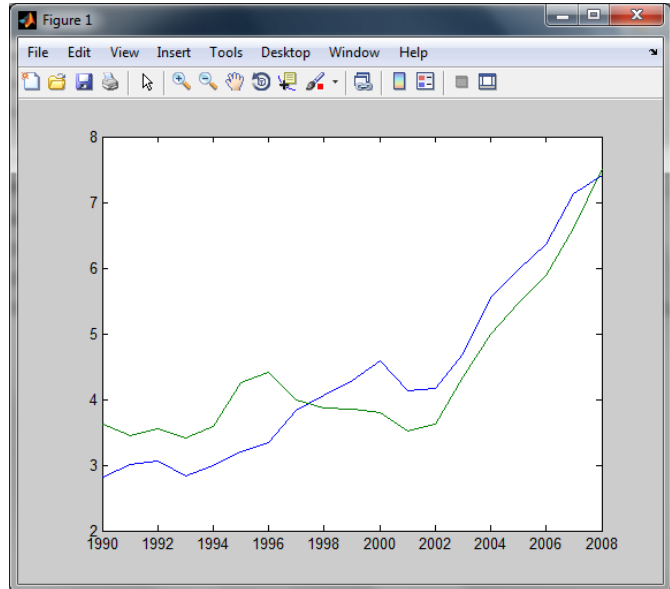
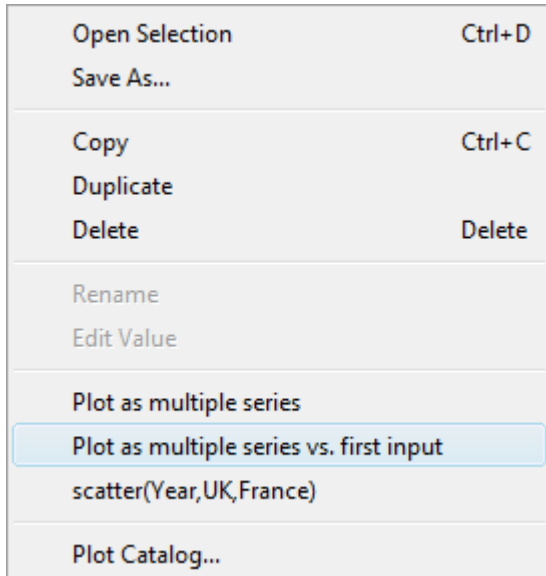
Command Window

Visualizing Data

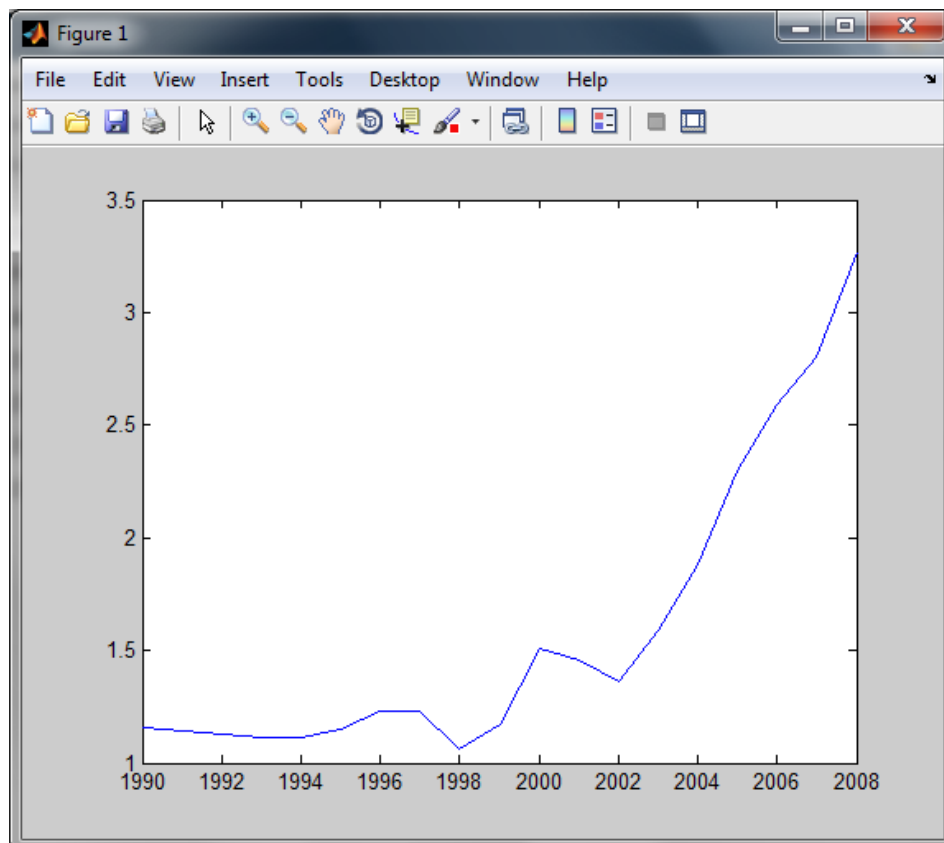
Matlab has a powerful set of tools that will help you display and visualize you data so you can discover trends and interesting changes. In order to plot the gas prices of the *UK* versus *years*, start by selecting the variable *Year* from the Workspace pane and then hold your keyboard *Ctrl* key as you click the *UK* variable in the *Workspace* pane. You should always select the variable you want to be plotted as the x-axis first and then select the variables for the y-axis. Right click on any of the selected variables and choose *plot(year, UK)*, see the figure window.



If you want to plot many variables versus the same axis (for example, plotting the *UK* and *France* gas prices versus years, start by selecting *year* and then hold down the *Ctrl* key as you select *UK* and *France*, then right click any of the variables and choose *plot as multiple series vs. first input*. See the resulting figure.

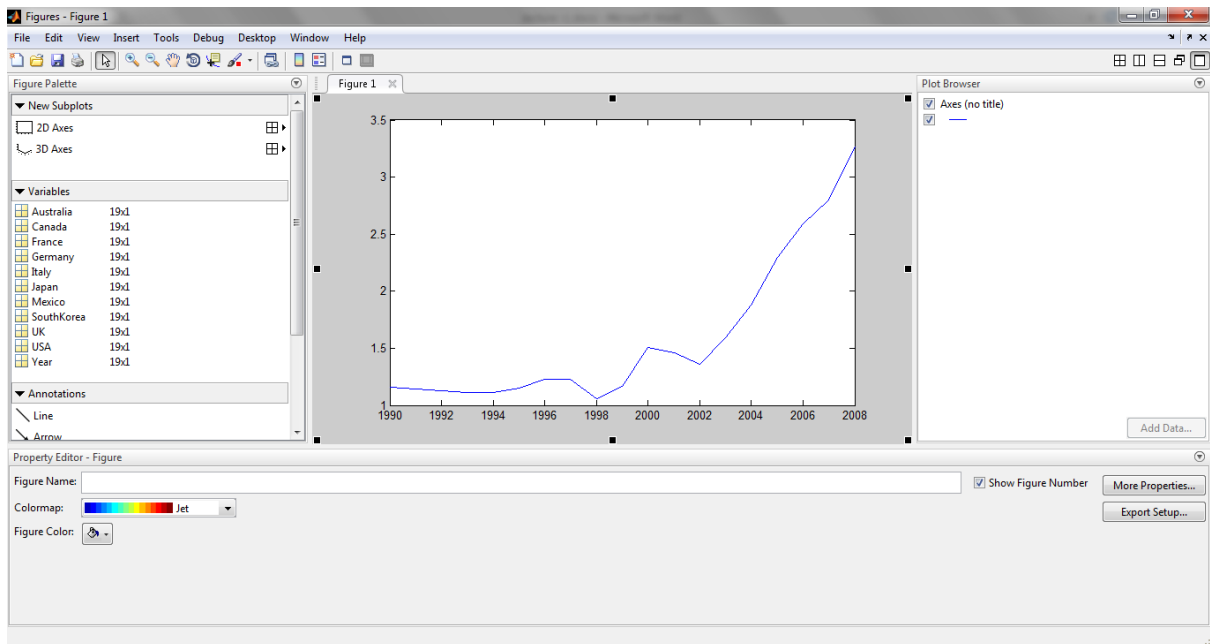


Quick Quiz: plot the USA gas prices vs year.
(your figure should look like the one below)

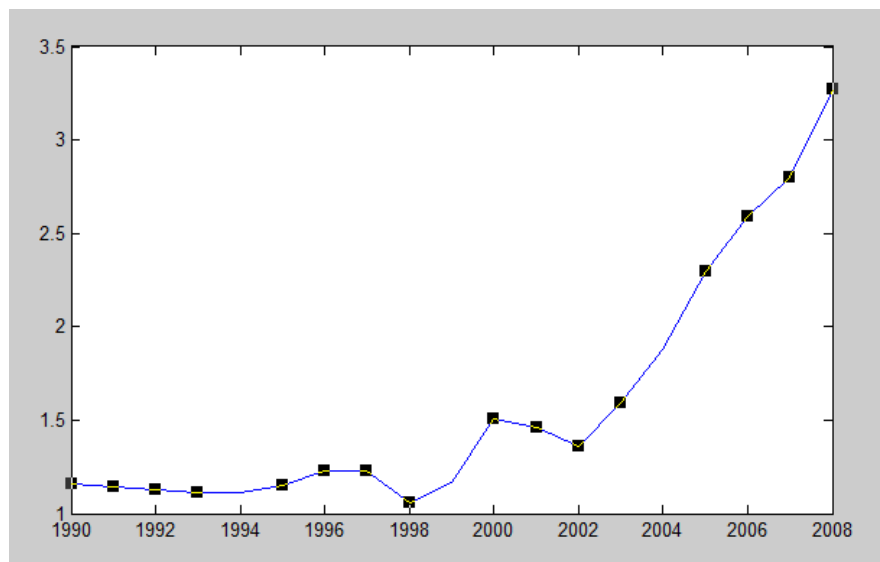


Formatting Plot

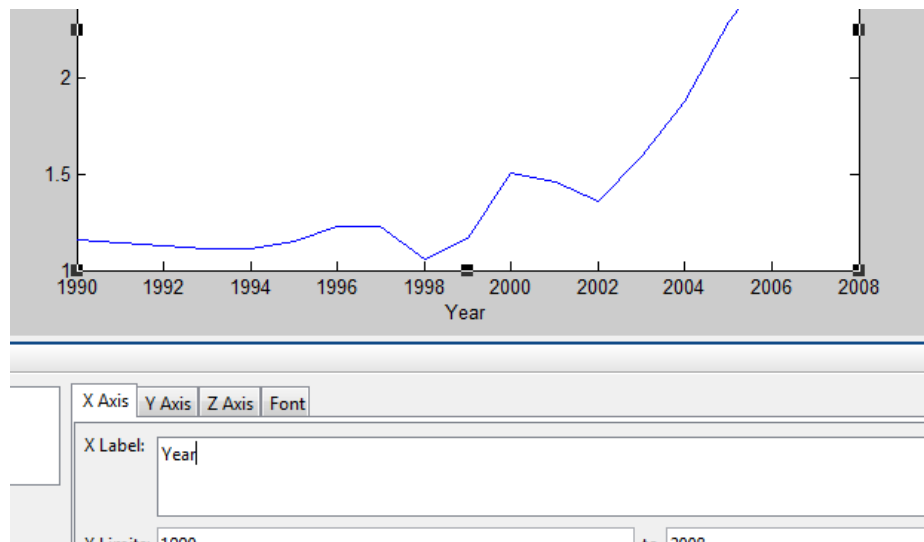
With the USA vs year figure active, click on the plot tools. See the following figure:



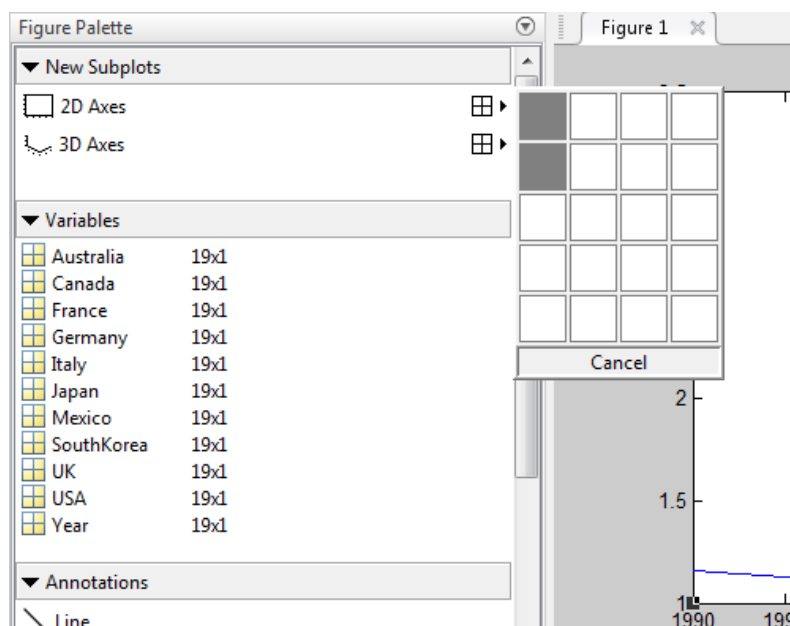
Every item in your plot has some properties. In order to change these properties, you need to select that item. Go ahead and click on the blue plot line. The line should become highlighted by little black squares.



In the property editor pane, you can change the line colour, its weight, marker, and many more. Click on any empty space inside the plot to change the properties of the axis. In the property editor, type *year* for the x-label and *USA* for the y-label.

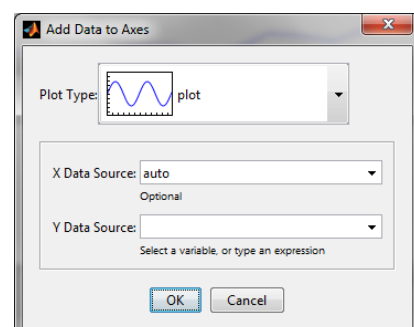


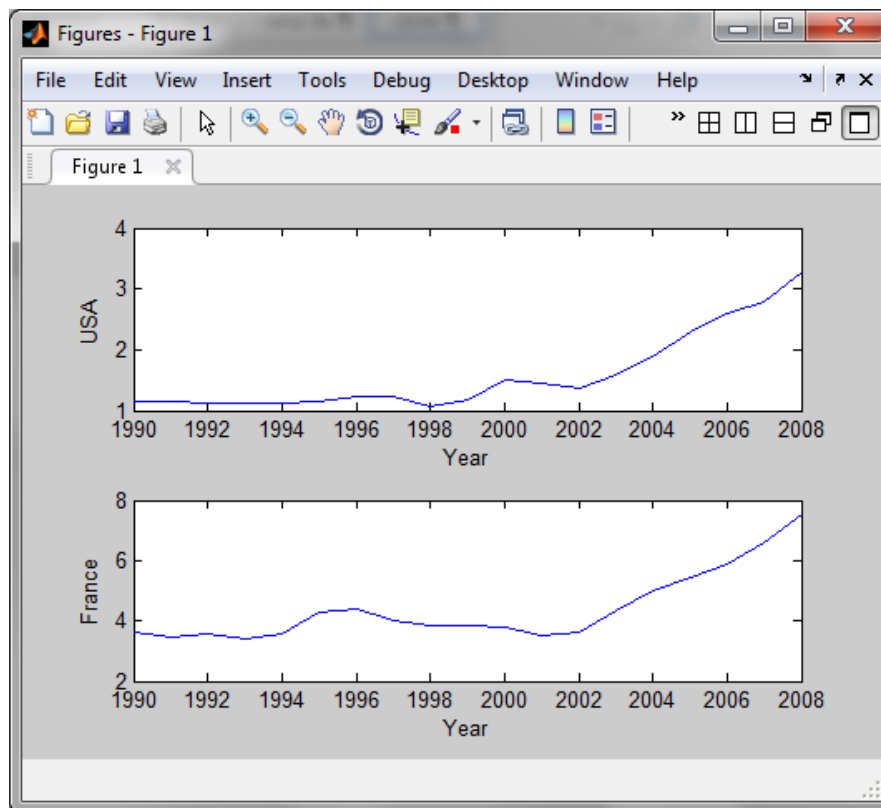
Now let us add a second graph to the same figure and let us make them arranged in a vertical fashion. For that, click the little arrow to the left of the *2D Axis* in the *New subplot* pane, and select two vertical boxes:



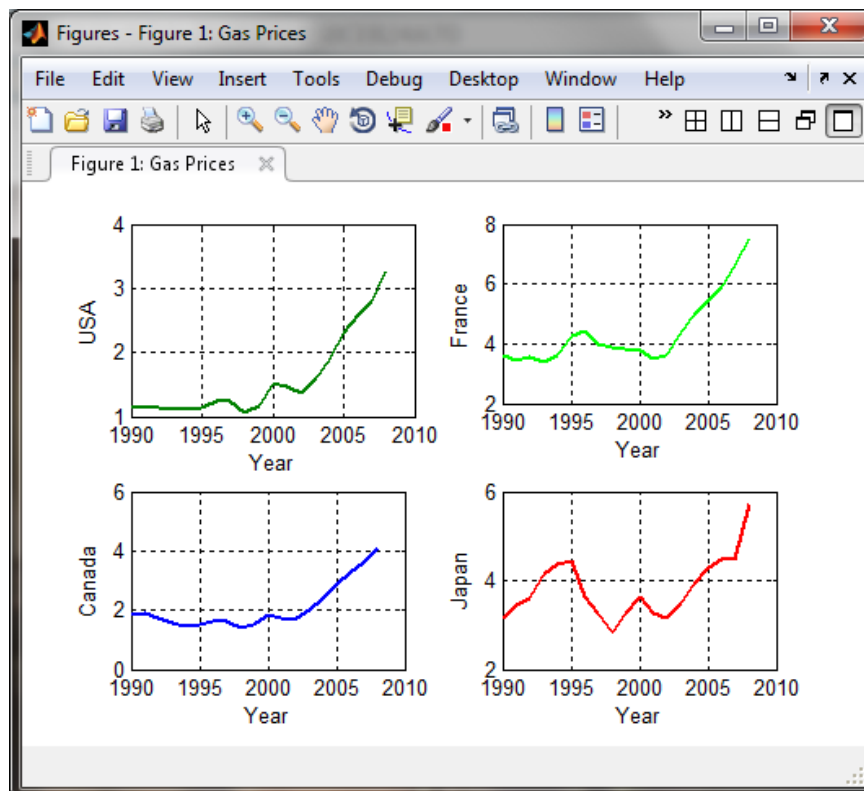
Select the new subplot and click on the add data button in the Plot browser pane. See the add data to Axes windows

For the *X data source*, select the year variable and for the *Y data source*, select France. Then fill the x and y label with year and France respectively. Finally click on the *hide the plot tools* button. Your end-result should look like the image down below



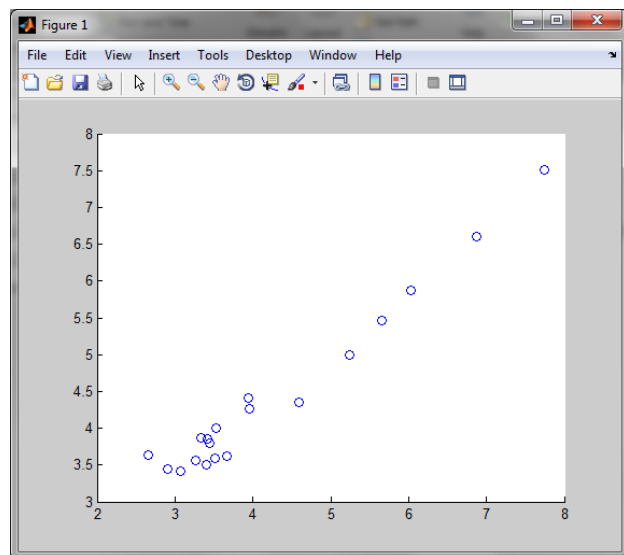
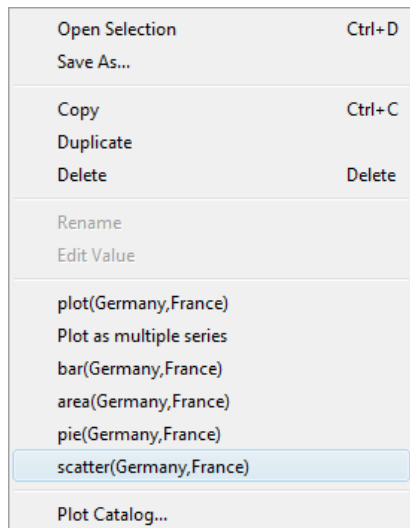


Quick Quiz. Try to replicate the following window using the plot tools property editor

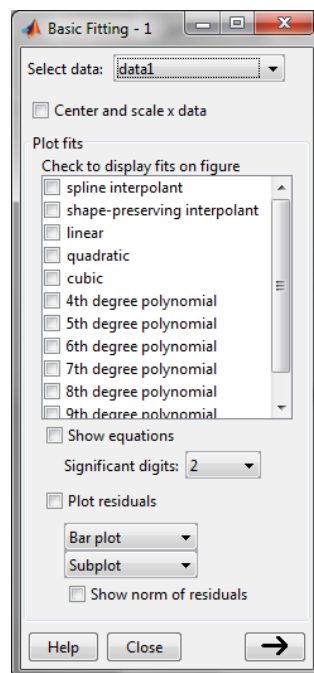


Basic Fitting

Visualizing data is important but we can do more! We can use regression to predict how the data behave by finding the formula that best describe our data. This is often applied on a scatter plot. Start by plotting the gas prices of Germany vs France. For that, select Germany and then hold down the Ctrl key while you select France, right click and choose *scatter(Germany,France)*



In figure 1 window, click on the tools menu and then select *basic fitting*. See the basic fitting window



In the basic fitting window choose linear for linear regression, click on the horizontal arrow. Matlab will display the equation that represent the relationship between the gas prices of Germany and France. Click the same horizontal arrow. Matlab should show you a box where you can enter values for the German gas prices and check the corresponding France prices based on linear formula we just found. Type 3.94 and click evaluate. The result should be 4.18. Comparing this value with the 4.41 actual gas price, proves that our approximation is very good.

Homework

Plot the gas prices of Mexico vs year and predict the gas prices in the year 2010. Repeat for Canada.